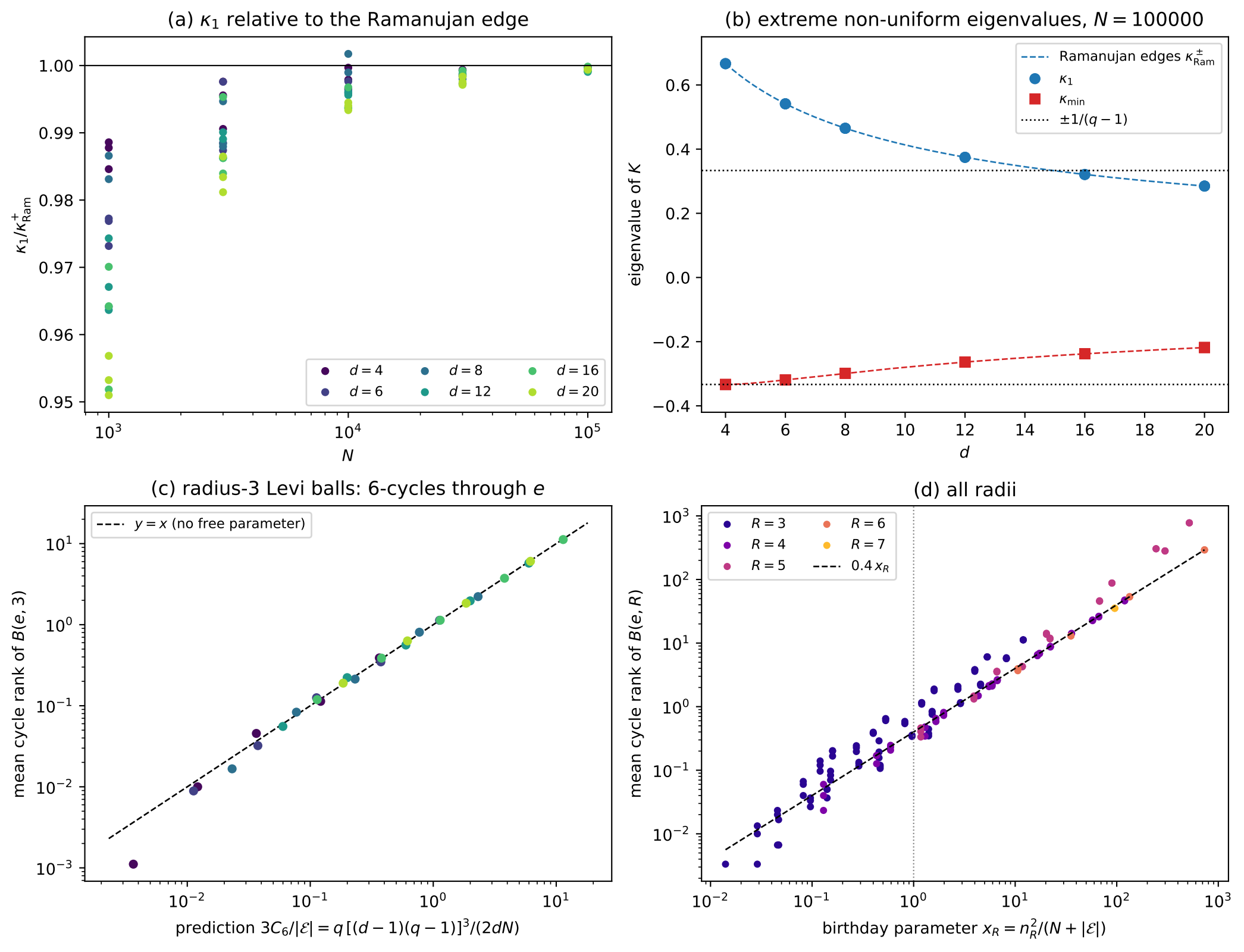
(a) κ_1 relative to the Ramanujan edge
$\kappa_1/\kappa^+_{\mathrm{Ram}}$
1.00
0.99
0.98
0.97
0.96
0.95
10^3
10^4
10^5
N
$d = 4$
$d = 6$
$d = 8$
$d = 12$
$d = 16$
$d = 20$
(b) extreme non-uniform eigenvalues, $N = 100000$
eigenvalue of K
0.6
0.4
0.2
0.0
−0.2
−0.4
4
6
8
10
12
14
16
18
20
d
Ramanujan edges $\kappa^{\pm}_{\mathrm{Ram}}$
κ_1
$\kappa_{\min}$
$\pm 1/(q-1)$
(c) radius-3 Levi balls: 6-cycles through e
mean cycle rank of $B(e, 3)$
$y = x$ (no free parameter)
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-2}
10^{-1}
10^0
10^1
prediction $3C_6/|\mathcal{E}| = q\,[(d-1)(q-1)]^3/(2dN)$
(d) all radii
mean cycle rank of $B(e, R)$
$R = 3$
$R = 4$
$R = 5$
$R = 6$
$R = 7$
$0.4\,x_R$
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-2}
10^{-1}
10^0
10^1
10^2
10^3
birthday parameter $x_R = n_R^2/(N + |\mathcal{E}|)$